Eight-Issuer Climate Scenario-to-Value Methodology

A transparent mathematical specification for transition risk, physical risk, macro/supply transmission, FCFF and discounted valuation

Document purpose	Explain the complete calculation without distributing implementation code
Coverage	AAPL, MSFT, CAT, DE, CVX, XOM, NEE and DUK
Model horizon	Annual paths from 2026 to 2050
Model release	CFP-2026-003-R17-20260804T234500Z
Prepared for	Academic and investment-team methodology review
Date	8 August 2026

Interpretation boundary
The headline output is a conditional target-versus-reference enterprise-value differential. It is not a forecast return, event probability, audited valuation, regulatory VaR, or reproduction of Bloomberg/MSCI proprietary models.

The document separates observed public evidence, scenario inputs, disclosed model assumptions, and derived outputs. Every DUK number shown below reconciles to the governed application export and the accompanying Excel review workbook.

Abstract

The model asks how the present value of an issuer changes when the assumed climate pathway changes. A target scenario and a reference scenario are evaluated using the same financial starting point, discount rate and terminal-growth rule. Three cash-flow channels are calculated separately: direct transition costs, physical costs, and macro/supply-chain costs. Annual target and reference FCFF paths are then discounted. Exact Shapley attribution allocates the total enterprise-value difference back to the three channels.

Most defensible use
Cross-issuer and cross-scenario comparison under common definitions, combined with sensitivity ranges and visible data-confidence labels. Stand-alone point estimates should receive less weight than direction, driver attribution, rank stability and data quality.

1. Calculation workflow

Stage	Calculation	Output used by an analyst
1. Scenario question	Select target s, reference r, horizon T and Low/Base/High assumption case	A clearly defined counterfactual question
2. Evidence layer	Load FCFF, emissions, PP&E;, facility/FEMA coverage and scenario paths	Observed-versus-assumed audit trail
3. Channel costs	Calculate Transition(t), Physical(t), and Macro/Supply(t)	Annual dollar pressure by driver
4. Scenario FCFF	Subtract the selected scenario costs from the same baseline FCFF path	Target and reference annual cash flows
5. DCF	Discount both paths using the same WACC and terminal-growth assumption	Target and reference enterprise values
6. Decision readout	Report relative value differential, channel attribution, assumption range and data confidence	Peer comparison, concentration review and issuer research

$$\Delta V_i(s,r) = V_{i,s} / V_{i,r} - 1 \quad (1)$$

A negative value means the target scenario produces a lower modeled enterprise value than the reference. The comparison is conditional: no probability is assigned to either scenario.

2. Data architecture and classification

The model uses a hierarchy rather than treating every number as equally reliable. Observed data enter without being relabeled as assumptions. Assumptions are stored by case. Derived outputs carry the release identifier and source classification.

Layer	Examples	Role in calculation	Current limitation
Observed issuer financials	SEC CFO, CapEx, net PP&E;, debt, tax	Normalize FCFF, measure fixed-asset exposure, estimate WACC inputs and CapEx/CFO dependency	Filing conventions and maintenance CapEx require judgment
Observed emissions	Company Scope 1, market-based Scope 2 and published Scope 3	Scale transition-cost channels	Scope 3 boundaries differ across issuers
Facility/hazard evidence	EPA parent facilities joined to FEMA county building EAL	Estimate acute physical exposure rate and coverage confidence	No asset value by facility; AAPL/MSFT have no governed issuer facilities
Scenario evidence	NGFS carbon-price paths and CIE USA labour-productivity pathways	Create target and reference paths	Some macro/supply scenario coefficients remain transparent prototype calibrations
Market inputs	Beta, market capitalization, Treasury risk-free rate	Estimate WACC	Public market fields are point-in-time snapshots, not licensed vendor histories
Model assumptions	Incidence, pass-through, recovery, labour conversion, macro elasticity and terminal growth	Convert scenario/evidence quantities into FCFF pressure	Not issuer-observed; handled through Low/Base/High cases

2.1 Physical-data confidence

Confidence is based on issuer facility-to-FEMA county match coverage: High above 80%, Medium from 50% to 80%, Low below 50%, and Unverified when no issuer facilities exist in the governed evidence set. Unverified is not zero risk.

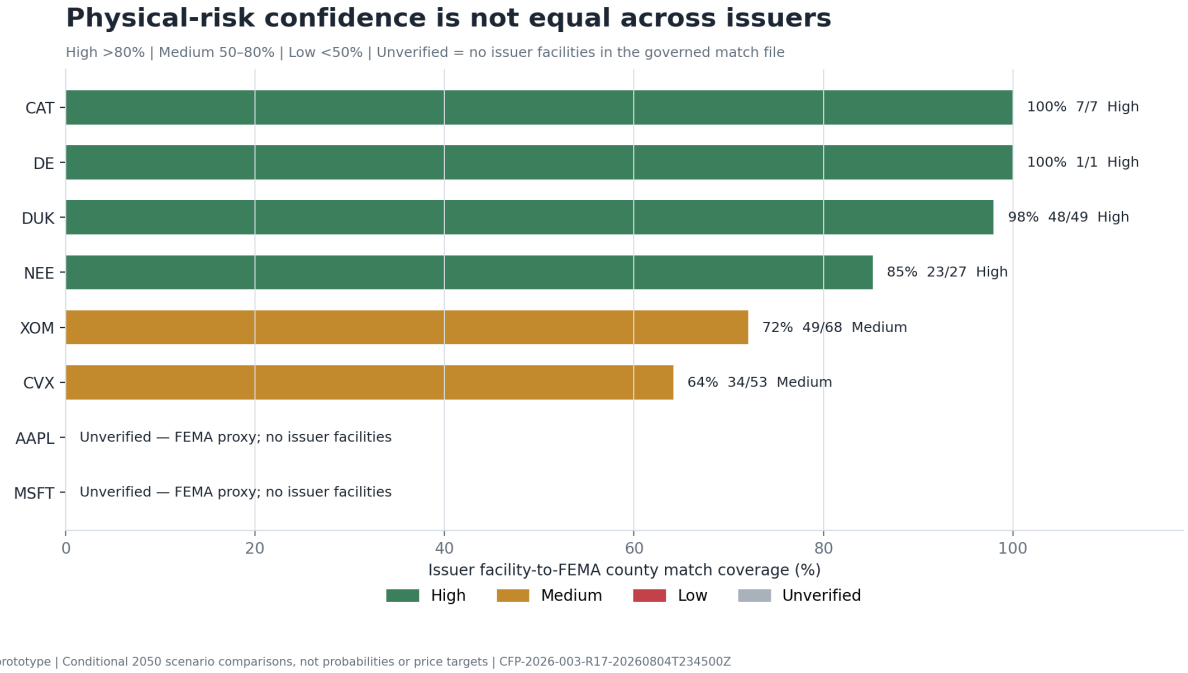


Figure 1. Facility evidence differs materially across the eight issuers; AAPL and MSFT remain explicitly Unverified.

3. Transition-risk calculation

Transition pressure is decomposed into three emissions scopes. Reported emissions are not treated as identical liabilities. Scope 1 receives direct policy coverage, Scope 2 receives purchased-energy incidence, and Scope 3 receives a smaller value-chain incidence. All three are reduced by the modeled emissions path and by cost pass-through.

$$E_{q,i,t} = E_{q,i,0}(1 - d_i)^t, \quad q = 1, 2, 3 \tag{2}$$

$$C_{1,i,s,t} = E_{1,i,t} \cdot P_{s,t} \cdot \kappa_{1,i} \cdot (1 - \pi_i) \tag{3}$$

$$C_{2,i,s,t} = E_{2,i,t} \cdot P_{s,t} \cdot \kappa_{1,i} \cdot \kappa_{2,i} \cdot (1 - \pi_i) \tag{4}$$

$$C_{3,i,s,t} = E_{3,i,t} \cdot P_{s,t} \cdot \kappa_{3,i} \cdot (1 - \pi_i) \tag{5}$$

$$TR_{i,s,t} = C_{1,i,s,t} + C_{2,i,s,t} + C_{3,i,s,t} \tag{6}$$

Here E is reported tCO2e, d is the annual emissions-decline assumption, P is the scenario carbon-price path, κ1 is Scope 1 policy coverage, κ2 is Scope 2 incidence, κ3 is Scope 3 value-chain incidence, and π is pass-through. The target-reference transition effect is TR(i,s,t) - TR(i,r,t). Atmospheric CO2 concentration is excluded from this equation because it is not a policy price.

3.1 DUK Base assumptions

Input	Value	Classification
Scope 1 / 2 / 3	79.756m / 0.254m / 27.608m tCO2e	Reported company evidence
Annual emissions decline d	3.0%	Model assumption
Scope 1 policy coverage κ1	75.0%	Model assumption
Scope 2 incidence κ2	50.0%	Model assumption
Scope 3 incidence κ3	0.5%	Sector-informed model assumption
Pass-through π	50.0%	Model assumption
2050 Current Policies carbon-price differential	$2.1579/tCO2	Governed scenario path differential

At 2050 these equations produce Scope 1 $30.1m, Scope 2 $47,991, and Scope 3 $69,550. Total terminal transition pressure is $30.3m.

4. Physical-risk calculation

Acute and chronic physical risks use different evidence. Acute risk applies a FEMA building expected-annual-loss rate to the economically covered share of SEC net PP&E.; Chronic risk applies the CIE labour-productivity pathway to the labour-sensitive share of terminal FCFF.

$$A_{i,s,t} = PP\&E_i \cdot \omega_i \cdot EAL_i \cdot (1 - \rho_i) \cdot a_s \cdot t/T \quad (7)$$

$$H_{i,s,t} = FCFF^0_{i,t} \cdot \ell_i \cdot \eta_i \cdot L_{s,t} \quad (8)$$

$$PR_{i,s,t} = A_{i,s,t} + H_{i,s,t} \quad (9)$$

PP&E; is SEC net property, plant and equipment; ω is covered PP&E; share; EAL is the emissions-weighted FEMA county building-loss rate; ρ is insurance/adaptation recovery; a is a scenario acute factor; ℓ is labour-cost share; η converts labour-productivity pressure into FCFF; and L is the CIE scenario labour-productivity-loss path.

DUK acute loss check
$129.959bn × 65% × 0.17566% × (1 - 50%) × 1.00 = $74.2m. The FEMA rate uses 48 matched counties from 49 governed facilities (97.959% coverage).

DUK chronic loss check
$2.513bn × 10% labour share × 25% FCFF conversion × 7.57987% = $4.8m.

5. Macro and supply-chain calculation

$$m_i = \text{clip}(0.50 + CapEx_i/CFO_i, 0.50, 2.00) \quad (10)$$

$$MS_{i,s,t} = FCFF^0_{i,t} \cdot (GDP_{s,T} + Supply_{s,T}) \cdot \varepsilon \cdot m_i \cdot t/T \quad (11)$$

The scenario elasticity ε is common within an assumption case; issuer differentiation comes from the SEC CapEx/CFO dependency scale m. For DUK Base, ε = 0.60, m = 1.6374 and the Current Policies terminal GDP-plus-supply stress is 2.0%. The resulting 2050 macro/supply pressure is $49.4m.

Important qualification
The CIE productivity path and FEMA EAL are observed scenario/hazard evidence. Covered PP&E;, recovery, labour conversion and macro elasticity are assumptions. The Current Policies GDP and supply terminal coefficients are transparent prototype scenario calibrations, not licensed Bloomberg/MSCI parameters.

6. From channel costs to FCFF and enterprise value

$$FCFF^{0}_{i,t} = FCFF_{i,2025}(1 + g_i)^{t-2025} \quad (12)$$

$$FCFF_{i,s,t} = \max[0, FCFF^{0}_{i,t} - TR_{i,s,t} - PR_{i,s,t} - MS_{i,s,t}] \quad (13)$$

$$V_{i,s} = \Sigma_{t=1}^{T} FCFF_{i,s,t}/(1+WACC_i)^t + FCFF_{i,s,T}(1+g_i)/[(WACC_i-g_i)(1+WACC_i)^T] \quad (14)$$

$$WACC_i = (E/V)(r_f + \beta_i ERP) + (D/V)r_d(1-\tau_i) \quad (15)$$

Both target and reference use the same normalized 2025 FCFF, WACC and terminal growth; only scenario-conditioned costs change. A zero FCFF floor prevents a stress from generating a negative enterprise value. When the floor binds, the output is labelled as a nonlinear boundary condition rather than an exact point Climate VaR.

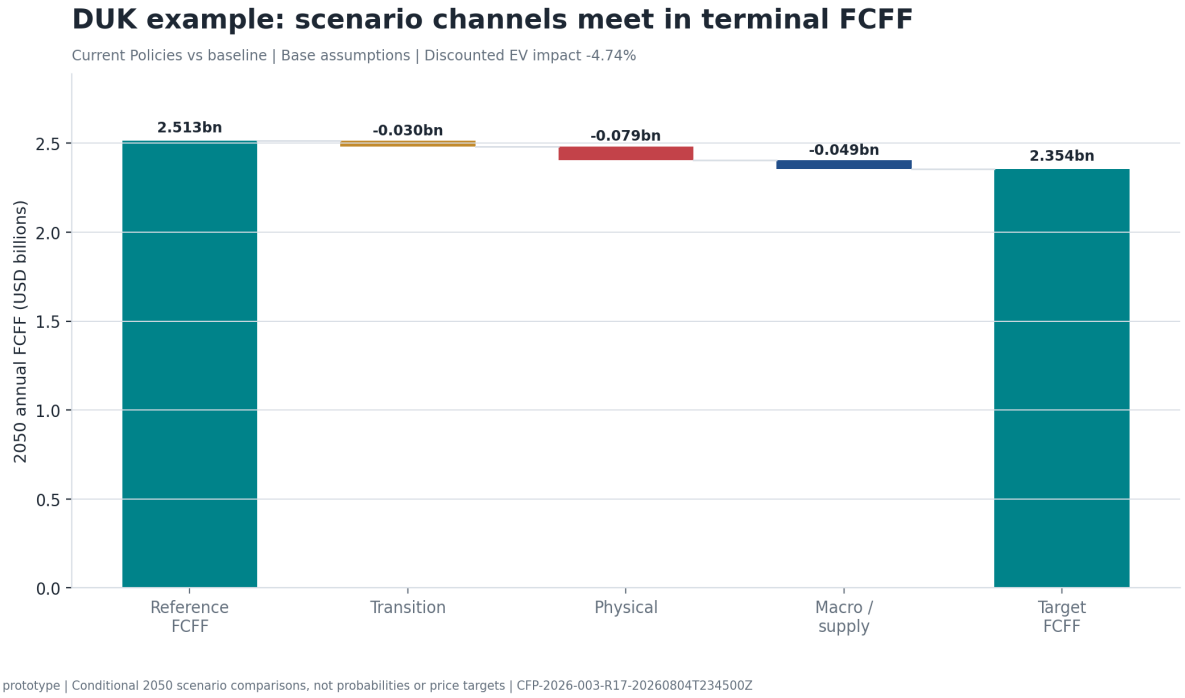


Figure 2. DUK 2050 terminal FCFF bridge for Current Policies versus the climate-agnostic baseline, Base assumptions.

DUK DCF quantity	Value
Normalized 2025 FCFF	$1.532bn
Reference terminal FCFF	$2.513bn
Target terminal FCFF	$2.354bn
WACC / terminal growth	5.116% / 2.00%
Reference / target enterprise value	$50.132bn / $47.758bn
Conditional value differential	-4.7356%

7. Attribution and scenario comparison

Because DCF is nonlinear, channel contributions are calculated with exact three-player Shapley values rather than by dividing terminal costs mechanically. For the channel set N = {transition, physical, macro/supply}, the contribution of channel j is:

$$\text{phi}_j = \text{sum}_{S \text{ in N without } j} |S|!(|N|-|S|-1)!/|N|! \cdot [v(S \text{ plus } j) - v(S)] \quad (16)$$

For DUK, transition contributes -0.8954%, physical -2.4384%, and macro/supply -1.4018%. The sum equals the total -4.7356% differential.

7.1 What each scenario comparison asks

Comparison	Investment question
Current Policies vs baseline	What is the broad modeled climate effect relative to a no-climate-cost counterfactual?
Net Zero vs baseline	Which issuers are most exposed to rapid transition pressure, allowing for lower physical damage?
Below 2°C vs Current Policies	Who benefits or suffers if policy tightens relative to the current-policy path?
Current Policies vs Delayed Transition	How much does the timing and concentration of transition pressure change value?
Fragmented World vs baseline	How exposed is the issuer to disorderly policy plus elevated physical and supply stress?

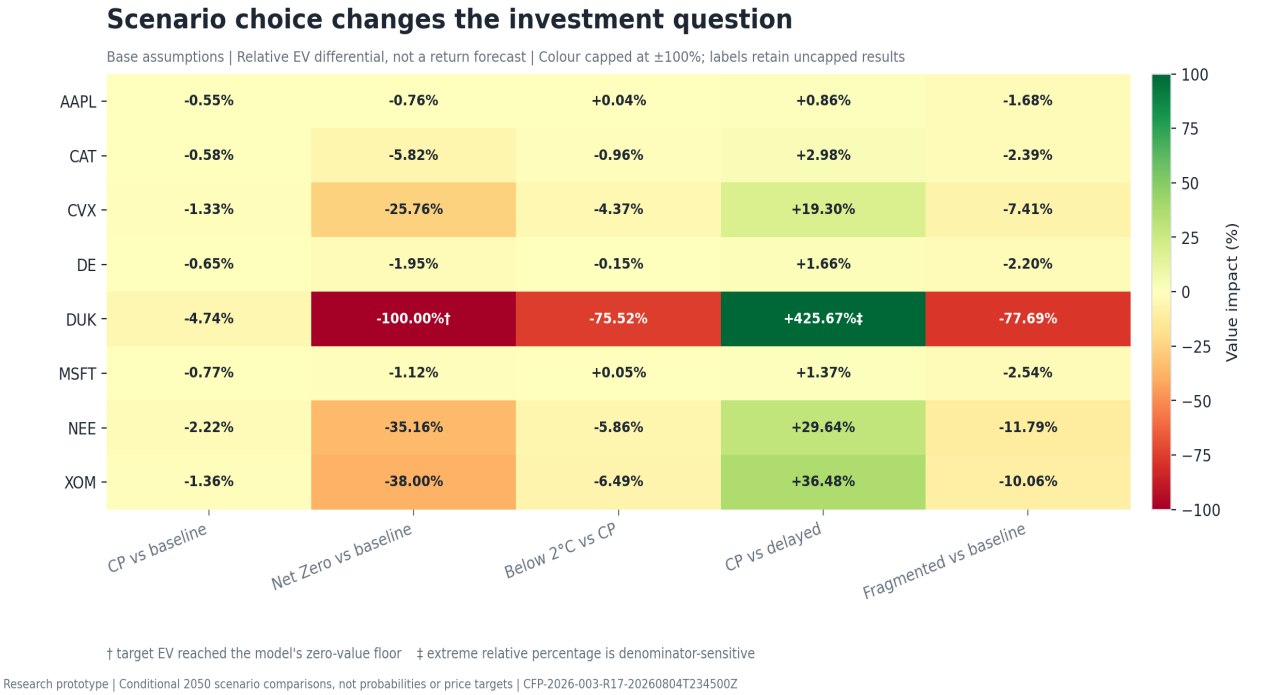


Figure 3. Eight issuers across five target/reference questions. Cell labels retain uncapped results; colour alone is capped at ±100% for readability.

8. How much should the numbers be trusted?

Use	Current confidence	Reason
Direction and driver attribution	Moderate	Observed issuer inputs and explicit formulas; strong dependence on financial-conversion assumptions remains
Peer comparison under one scenario	Moderate	Common scenario definitions improve comparability; issuer data coverage is uneven
Low/Base/High range	Moderate	Shows sensitivity to disclosed parameters but is not a statistical confidence interval
Stand-alone point Climate VaR	Low	No calibrated scenario probabilities, independent damage-function validation or full value-chain/facility economics
Automatic trading or price target	Not approved	The model is a long-horizon research overlay, not a return forecast

8.1 Engineering verification

The reporting release contains 120 calculable rows (8 issuers × 5 scenario comparisons × 3 assumption cases). Ninety-three automated tests pass. Controls include equal-scenario zero conservation, transition/physical/macro reconciliation, DUK terminal-FCFF tie-out, exact Shapley reconciliation, missing-versus-zero handling, facility-confidence classification, and explicit flags for enterprise-value floors and denominator-sensitive results. The Excel review workbook reports overall export status OK.

8.2 Extreme-value interpretation

DUK reaches the zero-value floor under Net Zero Base (-100%). This means modeled cash-flow stress is large enough to exhaust the target enterprise-value calculation under the stated assumptions; it is not evidence that the stock will lose exactly 100%. DUK Current Policies versus Delayed Transition produces +425.67% because the reference enterprise value becomes very small. This is a denominator-sensitive scenario differential, not a +425.67% forecast return.

Recommended communication
Use the model to explain why rankings change across scenarios, which channel dominates, and which missing data should be investigated. Do not ask a client to trust one percentage without its scenario pair, assumption case, data confidence and boundary flags.

9. Relationship to Bloomberg and MSCI

Dimension	This prototype	Commercial Climate VaR platforms
Relationship	Conceptual implementation of scenario-to-cash-flow-to-value logic	Proprietary production methodologies
Strength	Transparent formulas, public-source lineage, visible fallbacks and editable sensitivity cases	Broader issuer/facility/value-chain data, standardized benchmarking and institutional support
Weakness	Only eight governed demo issuers; several conversion parameters remain assumed; no independent investment validation	Less reproducible externally; exact formulas and calibrations may be proprietary
Replication claim	None	Not applicable

10. Parameter ledger and governance boundary

Parameter	DUK Base	Status / required improvement
Policy coverage	75%	Assumption; replace with jurisdiction and free-allocation mapping
Pass-through	50%	Assumption; estimate from rate cases, contracts and historical margins
Scope 2 / Scope 3 incidence	50% / 0.5%	Assumptions; calibrate from energy procurement and value-chain economics
Covered PP&E; / recovery	65% / 50%	Assumptions; require facility asset values, insurance and adaptation evidence
Labour share / FCFF conversion	10% / 25%	Assumptions; estimate from issuer cost structure and operating history
Macro elasticity	0.60	Common case assumption; estimate by sector and issuer
Macro dependency scale	1.6374×	Derived from SEC CapEx/CFO using a disclosed bounded rule
WACC / terminal growth	5.116% / 2.00%	Public financial inputs plus ERP/g assumptions; governance required
FCFF floor	Zero	Nonlinear boundary; floor-binding cases must remain flagged

Notation

Symbol	Meaning	Symbol	Meaning
i	Issuer	s, r	Target and reference scenarios
t, T	Year and selected horizon	E(q)	Reported emissions for Scope q
P(s,t)	Scenario carbon-price path	EAL	FEMA building expected-annual-loss rate
TR, PR, MS	Transition, physical and macro/supply costs	FCFF0	Baseline free cash flow to firm
V(i,s)	Enterprise value under scenario s	φ(j)	Shapley value contribution

References and source notes

1. Network for Greening the Financial System (NGFS), Phase V climate scenario framework and scenario data.
2. Climate Impact Explorer (CIE), USA GDP-weighted labour-productivity-loss pathways.
3. U.S. Securities and Exchange Commission, EDGAR/XBRL issuer financial statements.
4. U.S. Environmental Protection Agency, Greenhouse Gas Reporting Program facility records.
5. Federal Emergency Management Agency, National Risk Index county building expected annual loss.
6. U.S. Treasury, dated risk-free-rate evidence.
7. Issuer sustainability reports and emissions disclosures for Scope 1, Scope 2 and published Scope 3.
8. Public market-data snapshots used for beta and capitalization inputs.
9. Climate-Finance-Trading-Platform governed release CFP-2026-003-R17-20260804T234500Z and Eight_Issuer_Climate_Stress_Test.xlsx.

Document control
This document explains the governed research methodology and its current limitations. It intentionally omits source code, local file organization, credentials and implementation details. Numeric results reproduce the final 2050 export generated on 8 August 2026.